

Chapter 7 — LIKE Operator

7.1 Introduction

What is LIKE?

LIKE is used to search for patterns inside strings.

Unlike =, which requires an exact match, LIKE allows partial matching.

Example

```
SELECT * FROM Students WHERE Name = 'Aadya';
```

Only matches

Aadya

But

```
SELECT * FROM Students WHERE Name LIKE 'A%';
```

matches

Aadya
Arpit

__ Wildcard

Represents

Exactly one character.

Example

```
WHERE Name LIKE '_a%'
```

Matches

```
Raj
Ram
Harsh
```

```
WHERE Name LIKE 'A_____'
```

Matches exactly five-letter names starting with A. eg: Aadya

scaping Wildcards

Suppose your data actually contains

```
50%
```

Searching

```
WHERE Discount LIKE '50%'
```

means

```
Starts with 50
```

not

```
Contains the literal %
```

To search for a real %, escape it.

MySQL

```
WHERE Discount LIKE '50\%' ESCAPE '';
```

Similarly,

To search for an underscore

```
WHERE Code LIKE 'AB\_12' ESCAPE '';
```

Index Considerations

This is an important interview topic.

Suppose an index exists on

Name

Starts With

```
WHERE Name LIKE 'A%'
```

The index can usually be used because the database knows where names starting with A begin in the index.

Ends With

```
WHERE Name LIKE '%A'
```

Usually cannot efficiently use a normal B-tree index because the beginning of the string is unknown.

ILIKE (Case-Insensitive Search)

Important: ILIKE is not supported in MySQL. It is a PostgreSQL-specific operator. Different databases treat string comparisons differently.

In PostgreSQL:

```
WHERE Name LIKE 'aadya'
```

is case-sensitive.

It will not match

Aadya

To perform a case-insensitive search, PostgreSQL provides

ILIKE

Syntax

PostgreSQL

```
SELECT * FROM Students WHERE Name ILIKE 'aadya';
```

Matches

Aadya
AADYA
aadya
AaDyA

```
WHERE LOWER(Name) LIKE 'a%';
```

Works in MySQL, PostgreSQL, SQL Server, and Oracle